Al-Balqa Applied University

جامعة البلقاء التطبيقية

Prob. 1: For the questions 1,2,3,4,5 consider the circuit shown in figure 1 and answer (use nodal analysis method):

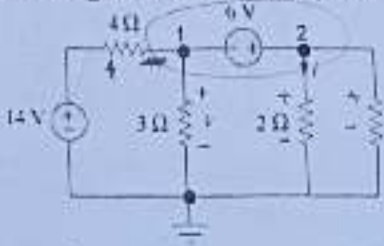


Fig.1

1) The value of V is: -

a) -200mV b) 200mV

c) 800mV [✓ d) 400mV] ✓

2) The value of the current (i) is:

a) 2.8A [✓ b) 1.4A]

c) -0.7A d) 0.7A

3) The power absorbed by 6 Ω resister is: ✓

a) -10.46W b) 10.46W [✓ c) 5.23W] d) 6.46W

4) The power absorbed by 6V voltage source is:

[✓ a) -22.4W] b) 44.8W c) 11.2W d) -44.8W

5) Write the required equations (do not simplify)

$$\frac{V_1-14}{4} = \frac{V_1}{3} + \frac{V_2}{6} + \frac{V_2}{2}$$

$$V_2 - V_1 = 6.$$

$$i = \frac{V}{2}.$$

Prob. 2: Consider the circuit shown in figure 2. Use superposition theorem and answer:

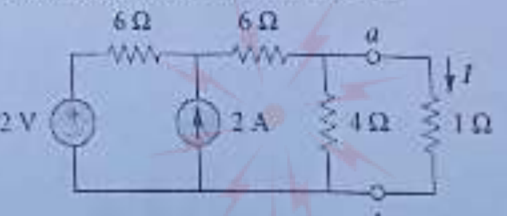


Fig. 2

a. The current I due to the 12V voltage source is:

$I' = 1.277\,A$

b. The current I due to the 2A current source is:

$I'' = 1.3\bar{3}\,A.$

c. The current I due to both sources is:

$I = I' + I'' = 2.6\,A.$

Prob.3 : (a) The charge transferred through a circuit element is given by $q(t)=4t^2+3t$ C. Calculate the current entering the element at instant $t=4$ s.

$i(4) = 35A.$

(b) A load draws 500 W when connected to a 230 V line. Calculate - the energy consumed in 3 hours and the conductance of the load?

$W = 5.4\,MJ.$

(c) Refer to figure 3; write the mesh equations in matrix form:

$$\begin{bmatrix} 9 & -2 & -2 & 0 & 0 \\ -2 & 10 & -4 & -1 & -1 \\ -2 & -1 & 4 & 0 & 0 \\ 0 & -1 & 0 & 8 & -3 \\ 0 & -1 & 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ -6 \\ 0 \\ -6. \end{bmatrix}$$

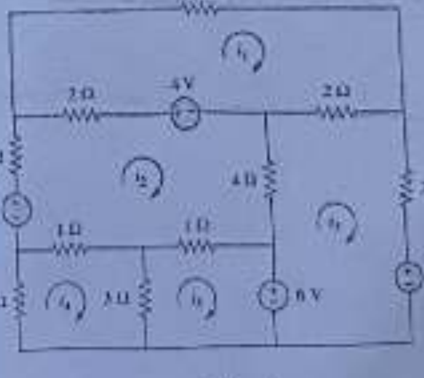
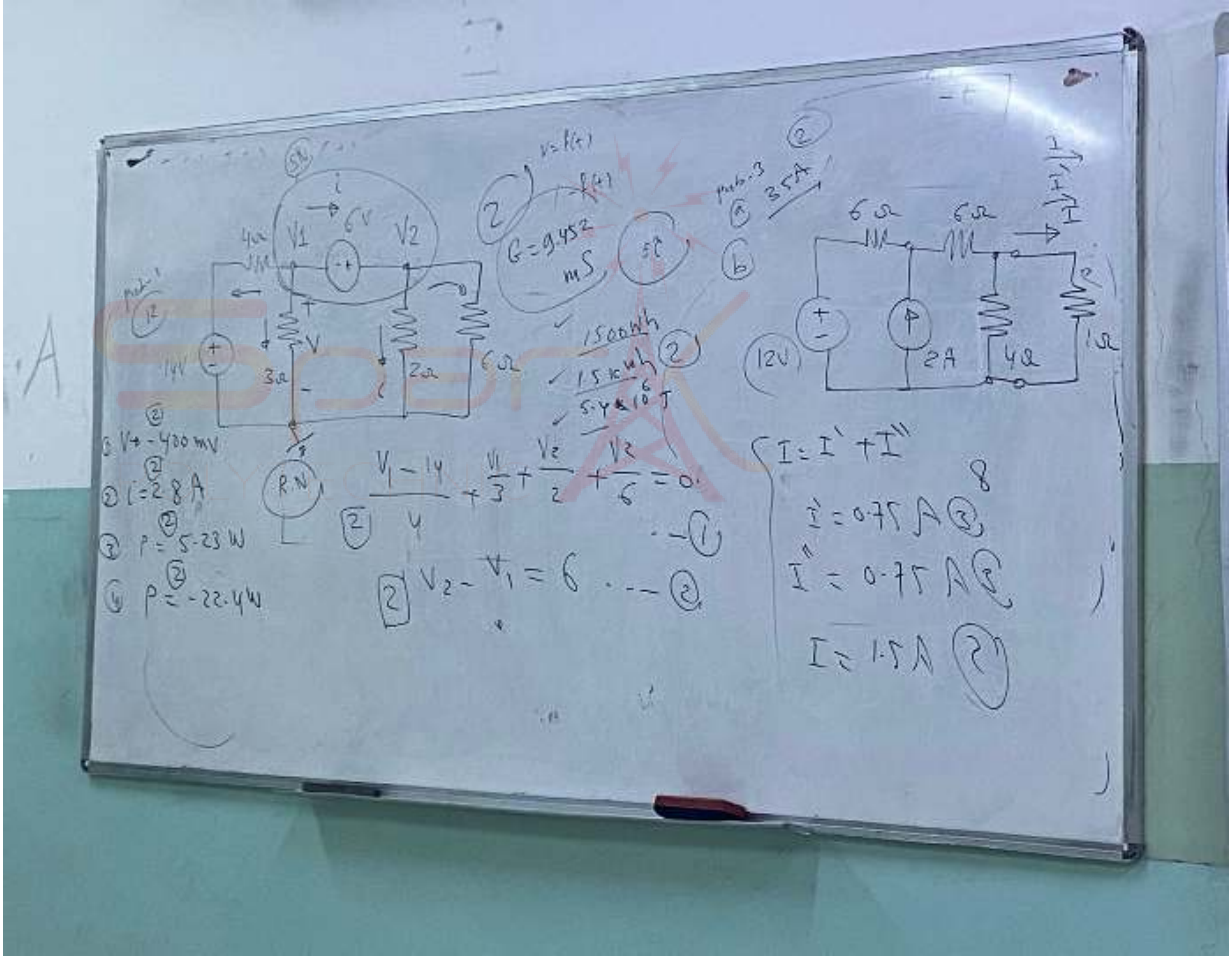


Fig 3.

Good luck



① V = −400 mV ②

② i = 2.8 A ②

③ P = 5.23 W ②

④ P = −22.4 W ②

$$\frac{V_1-14}{4}+\frac{V_1}{3}+\frac{V_2}{2}+\frac{V_2}{6}=0 \quad \ldots ①$$ ②

$$V_2-V_1=6 \quad \ldots ②$$ ②

$v = f(t)$

$i = f(t)$

② G = 9.452 mS

5C

1500 Wh

1.5 kWh ②

5.4×10^6 J

prob 3 ⓐ 35A ②

ⓑ

$I = I' + I''$ 8

$I' = 0.75$ A ②

$I'' = 0.75$ A ②

$I = 1.5$ A ②

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